# 풀이: 실험 문제 1

| 과제 1 | | 점수 |
|---|---|---|

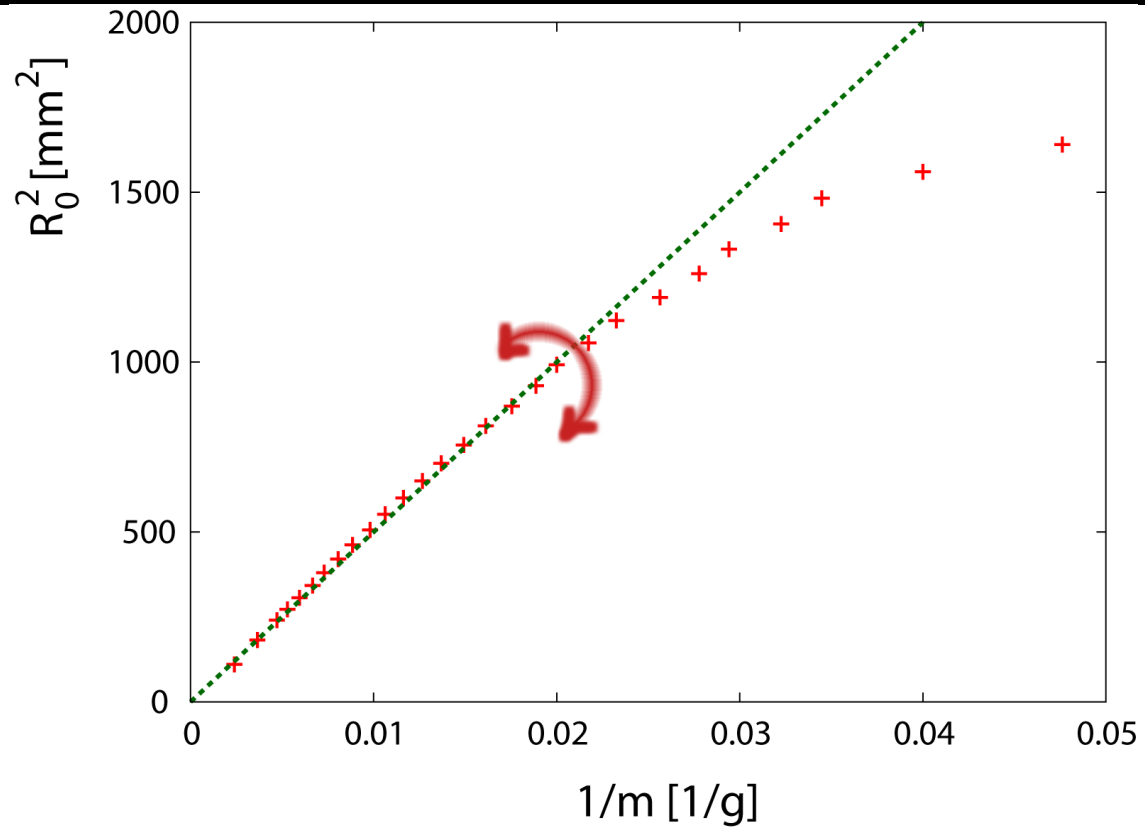
(a)

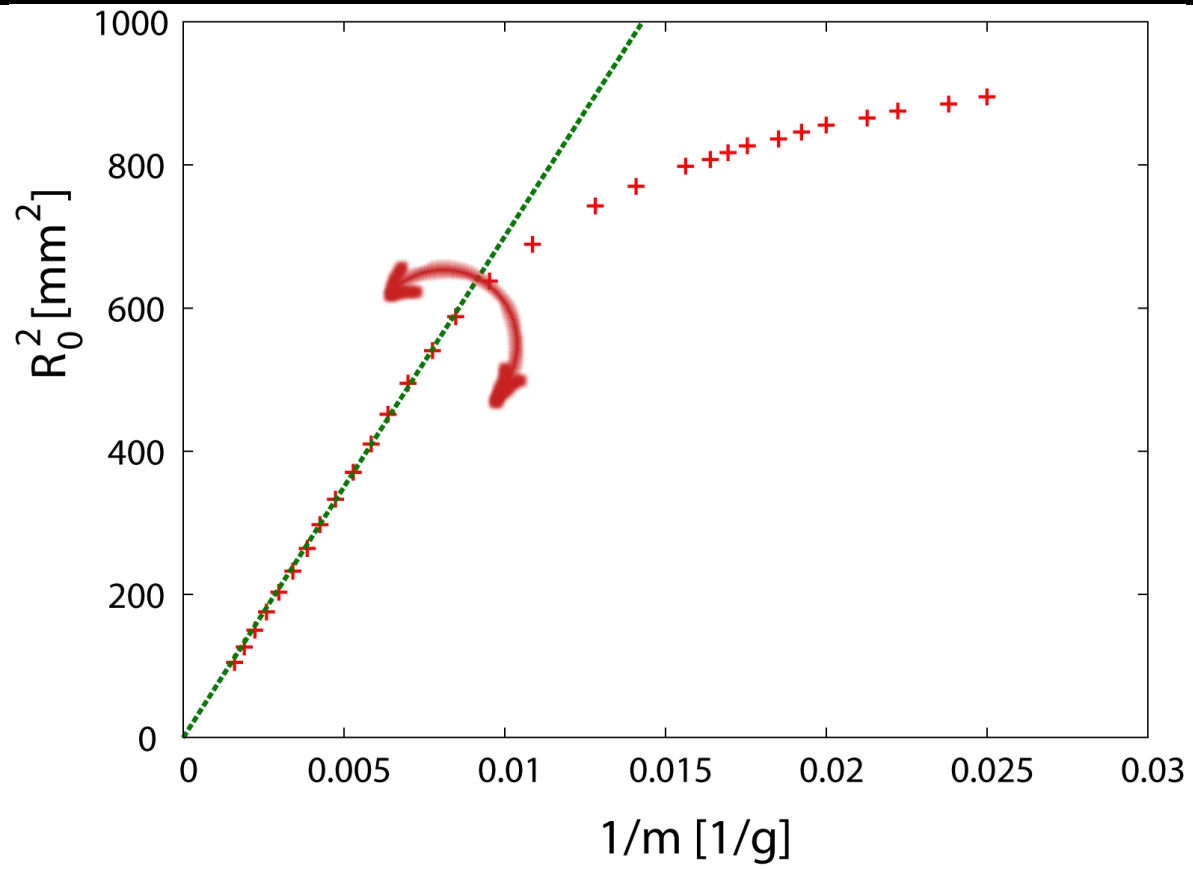
| m[g] | $R_0$[mm] |
|---|---|
| 21 | 40.5 |
| 25 | 39.5 |
| 29 | 38.5 |
| 31 | 37.5 |
| 34 | 36.5 |
| 36 | 35.5 |
| 39 | 34.5 |
| 43 | 33.5 |
| 46 | 32.5 |
| 50 | 31.5 |
| 53 | 30.5 |
| 57 | 29.5 |
| 62 | 28.5 |
| 67 | 27.5 |
| 73 | 26.5 |
| 79 | 25.5 |
| 86 | 24.5 |
| 94 | 23.5 |
| 102 | 22.5 |
| 113 | 21.5 |
| 124 | 20.5 |
| 137 | 19.5 |
| 150 | 18.5 |
| 168 | 17.5 |
| 189 | 16.5 |
| 212 | 15.5 |
| 274 | 13.5 |
| 417 | 10.5 |

0.95

| 과제 1 | | 점수 |
|---|---|---|
| (a) | | 0.95 |

| m[g] | $R_0$[mm] |
|---|---|
| 40 | 29.9 |
| 42 | 29.8 |
| 45 | 29.6 |
| 47 | 29.4 |
| 50 | 29.3 |
| 52 | 29.1 |
| 54 | 28.9 |
| 57 | 28.8 |
| 59 | 28.6 |
| 61 | 28.4 |
| 64 | 28.3 |
| 71 | 27.8 |
| 78 | 27.3 |
| 92 | 26.3 |
| 105 | 25.3 |
| 118 | 24.3 |
| 129 | 23.3 |
| 143 | 22.3 |
| 157 | 21.3 |
| 171 | 20.3 |
| 189 | 19.3 |
| 211 | 18.3 |
| 235 | 17.3 |
| 259 | 16.3 |
| 293 | 15.3 |
| 336 | 14.3 |
| 386 | 13.3 |
| 449 | 12.3 |

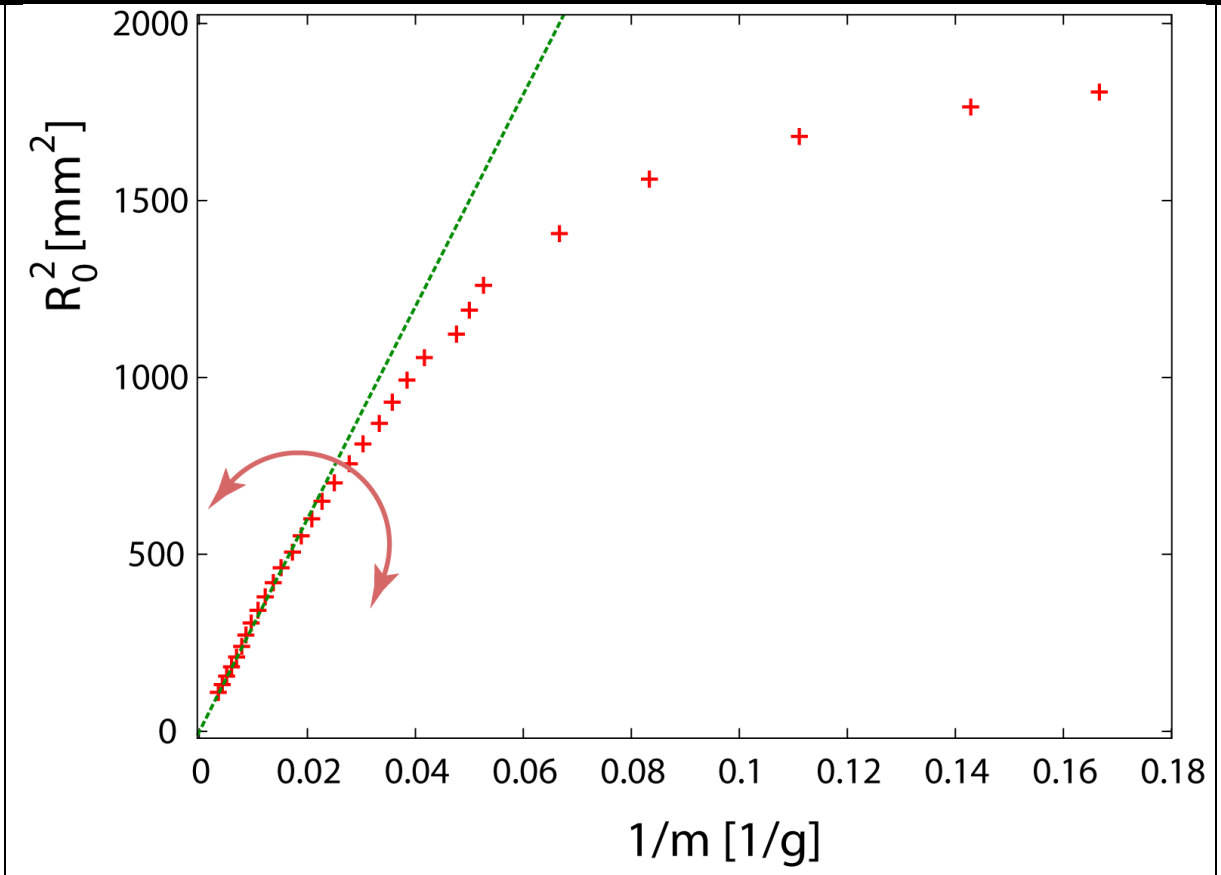
| 과제 1 | | 점수 |
|---|---|---|
| (b) |  <br>$a = 50000\ \text{g mm}^2$ | 1.4 |

| 과제 1 | | 점수 |
|---|---|---|
| (b) |  <br>$a = 70000\ g\ mm^2$ | 1.4 |

| 과제 1 | | 점수 |
|---|---|---|
| (b) | $\kappa = \frac{2ag}{\pi l} = 1.5\ \text{mJ}$ | 0.5 |
| | $\kappa = \frac{2ag}{\pi l} = 1.5\ \text{mJ}$ | 0.5 |
| | $\frac{R_0}{R_c} \le 0.70$ $\quad\quad$ $\frac{R_0}{R_c} \le 0.77$ | 0.5 |

| 과제 2 | | 점수 |
|---|---|---|
| | | 0.9 |

| m[g] | $R_0$[mm] |
|---|---|
| 6 | 42.5 |
| 7 | 42. |
| 9 | 41. |
| 12 | 39.5 |
| 15 | 37.5 |
| 19 | 35.5 |
| 20 | 34.5 |
| 21 | 33.5 |
| 24 | 32.5 |
| 26 | 31.5 |
| 28 | 30.5 |
| 30 | 29.5 |
| 33 | 28.5 |
| 36 | 27.5 |
| 40 | 26.5 |
| 44 | 25.5 |
| 48 | 24.5 |
| 53 | 23.5 |
| 58 | 22.5 |
| 66 | 21.5 |
| 73 | 20.5 |
| 82 | 19.5 |
| 92 | 18.5 |
| 104 | 17.5 |
| 116 | 16.5 |
| 127 | 15.5 |
| 145 | 14.5 |
| 168 | 13.5 |

| 과제 2 | | 점수 |
|---|---|---|
| |  <br>$a = 27000\ g\ mm^2$ | 0.9 |

| 과제 2 | | 점수 |
|---|---|---|
| | $\kappa$ =0.8 mJ | 1.0 |
| 과제 3 | | 점수 |
| | 파란 필름의 영률:<br>Y=2.0 GPa Y=2.0 GPa | 0.6 |
| | 무색 필름의 영률:<br>Y=2.5 GPa | 0.4 |
| 합계: | | 10 |